Stochastic Processes — 1060429

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Throughout, a Brownian motion (BM) is standard ($B_0 = 0$) with continuous paths, and $\mathcal{F}_t$ denotes the filtration in use. Apparent typos are pointed out where they occur.

Exercise 1

Relations between stochastic processes

1. Let $\Omega_0 = \{X, Y \text{ right-continuous}\} \cap \bigcap_{q\in\mathbb{Q}_+}\{X_q = Y_q\}$. The first event contains a measurable set of probability 1, and each $\{X_q = Y_q\}$ has probability 1 (modification), so Ω_0 contains a measurable set of probability 1 (countable intersection). For $\omega \in \Omega_0$ and $t \geq 0$ take $q_n \in \mathbb{Q}_+$, $q_n \downarrow t$; then $X_t(\omega) = \lim X_{q_n}(\omega) = \lim Y_{q_n}(\omega) = Y_t(\omega)$. Hence $X_t = Y_t$ for all t on Ω_0, i.e. X, Y are indistinguishable.

2. Let $Q_0 = \mathbb{Q} \cap [0, t_0)$ and

$$C_{t_0} = \bigcap_{k\geq 1}\bigcup_{m\geq 1}\bigcap_{s,u\in Q_0,\ |s-u|<1/m} \{|X_s - X_u| \leq 1/k\} \in \sigma(X_t : t \leq t_0).$$

Claim. If the path $X(\omega)$ is RCLL on $[0, t_0]$, then $\omega \in A_{t_0} \iff \omega \in C_{t_0}$.

If $X(\omega)$ is continuous on $[0, t_0)$ and has a left limit at t_0, it extends continuously to $[0, t_0]$, so it is uniformly continuous on $[0, t_0)$ and $\omega \in C_{t_0}$. Conversely, let $\omega \in C_{t_0}$, $k \geq 1$, and m as in the definition. For $s, u \in [0, t_0)$ with $|s - u| < 1/m$ choose $q_j, r_j \in Q_0$ with $q_j \downarrow s$, $r_j \downarrow u$ and $|q_j - r_j| < 1/m$ (possible as $s, u < t_0$). By right-continuity $|X_s - X_u| = \lim |X_{q_j} - X_{r_j}| \leq 1/k$. So $X(\omega)$ is uniformly continuous on $[0, t_0)$, hence $\omega \in A_{t_0}$.

(a) If every path is RCLL, the claim gives $A_{t_0} = C_{t_0} \in \sigma(X_t : t \leq t_0)$.

(b) Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, $\mathbb{P} = \text{Leb}$, and let C be the Cantor set. Since $|\mathcal{P}(C)| = 2^{\mathfrak{c}} > \mathfrak{c} = |\mathcal{B}([0, 1])|$, there is $E \subset C$ with $E \notin \mathcal{F}$. Define $X_t(\omega) = \mathbf{1}_E(\omega)\mathbf{1}_{\{t=\omega\}}$. Then $X_t = \mathbf{1}_{E\cap\{t\}}$ is the indicator of the empty set or of a single point, so it is measurable. For $\omega \notin E$ the path is $\equiv 0$. For $\omega \in E$ the path has a spike at $t = \omega$ and is not right-continuous there. Thus $\{X \text{ RCLL}\} = \Omega \setminus E \supseteq \Omega \setminus C$ and $\mathbb{P}(\Omega \setminus C) = 1$, so X is RCLL with probability 1. But $A_2 = \Omega \setminus E \notin \mathcal{F}$, and in particular $A_2 \notin \sigma(X_t : t \leq 2) \subseteq \mathcal{F}$.

(c) There is $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$ such that every path outside N is RCLL. By the claim, $A_{t_0} \triangle C_{t_0} \subseteq N$. Completeness of $\mathcal{F}_{t_0}$ (it contains every subset of a $\mathbb{P}$-null set) gives $A_{t_0} \setminus C_{t_0}$, $C_{t_0} \setminus A_{t_0} \in \mathcal{F}_{t_0}$. Hence $A_{t_0} = (C_{t_0} \setminus (C_{t_0} \setminus A_{t_0})) \cup (A_{t_0} \setminus C_{t_0}) \in \mathcal{F}_{t_0}$.

Brownian motion

1.

Remark. The condition should read $\mathbb{E}[|X_t - X_s|^\alpha] \leq M|t - s|^{1+\beta}$ for all $t, s \in [0, T]$. In (c) the bound should be $2^{-\gamma n}$, not 2^{-m}. In (e),(f) the index set is $[0, 1]$.

(a) Put $\tilde{X}_u = X_{uT}$, $u \in [0, 1]$. Then $\mathbb{E}|\tilde{X}_u - \tilde{X}_v|^\alpha \leq MT^{1+\beta}|u - v|^{1+\beta}$, which is the hypothesis on $[0, 1]$ with constant $MT^{1+\beta}$. If $\tilde{Y}$ is a modification of $\tilde{X}$ that is a.s. γ-Hölder, then $Y_t = \tilde{Y}_{t/T}$ satisfies $\mathbb{P}(Y_t = X_t) = \mathbb{P}(\tilde{Y}_{t/T} = \tilde{X}_{t/T}) = 1$, and $|Y_t - Y_s| \leq K|t/T - s/T|^\gamma = KT^{-\gamma}|t - s|^\gamma$.

(b) By Markov's inequality, $\mathbb{P}(|X_t - X_s| > \varepsilon) \leq \varepsilon^{-\alpha} M|t - s|^{1+\beta} \to 0$ as $t \to s$.

(c) By a union bound and Markov's inequality,

$$\mathbb{P}\Big(\max_{1\leq k\leq 2^n} |X_{k2^{-n}} - X_{(k-1)2^{-n}}| > 2^{-\gamma n}\Big) \leq 2^n \cdot 2^{\gamma\alpha n} M 2^{-n(1+\beta)} = M 2^{-n(\beta-\alpha\gamma)},$$

which is summable since $\gamma < \beta/\alpha$. By Borel–Cantelli, a.s. only finitely many of these events occur. Let $N(\omega)$ be the last n at which one occurs ($N = 0$ if none occurs). Then $\mathbb{P}(N < \infty) = 1$ and $\max_k |X_{k2^{-n}} - X_{(k-1)2^{-n}}| \le 2^{-\gamma n}$ for all $n > N(\omega)$.

(d) Fix ω with $N(\omega) < \infty$. *Claim:* for $m > n \ge N(\omega)$ and $s < t$ in D_m with $t - s < 2^{-n}$,

$$|X_t - X_s| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j}.$$

We use induction on m. For $m = n + 1$ we have $t - s = 2^{-(n+1)}$, and the bound holds by (c). For the step $m \to m + 1$, let $s, t \in D_{m+1}$. Let $s' = \min\{r \in D_m : r \ge s\}$ and $t' = \max\{r \in D_m : r \le t\}$. Then $s' - s,\ t - t' \in \{0, 2^{-(m+1)}\}$, and $s' \le t'$ (otherwise $s < t < s' \le s + 2^{-(m+1)}$, which is impossible in D_{m+1}). Since $t' - s' \le t - s < 2^{-n}$, the induction hypothesis (or $t' = s'$) and (c) at level $m + 1 > N$ give

$$|X_t - X_s| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j} + 2 \cdot 2^{-\gamma(m+1)}.$$

Now let $s < t$ in $D = \bigcup_m D_m$ with $t - s < 2^{-N(\omega)}$. Choose n with $2^{-(n+1)} \le t - s < 2^{-n}$, so $n \ge N(\omega)$, and choose $m > n$ with $s, t \in D_m$. Then

$$|X_t - X_s| \le 2 \sum_{j>n} 2^{-\gamma j} = \frac{2 \cdot 2^{-\gamma(n+1)}}{1 - 2^{-\gamma}} \le C_\gamma |t - s|^\gamma, \qquad C_\gamma = \frac{2}{1 - 2^{-\gamma}}.$$

(e) Let $s < t$ in D with $t - s \ge 2^{-N}$. The points of $D_{N+1} \cap (s, t)$ together with s, t form a chain $s = u_0 < \cdots < u_k = t$ in D with consecutive gaps $< 2^{-N}$ and $k \le 2^{N+1} + 1$. Each gap contributes at most $C_\gamma 2^{-N\gamma} \le C_\gamma |t - s|^\gamma$. Hence on $\{N < \infty\}$,

$$|X_t - X_s| \le K(\omega)|t - s|^\gamma \text{ for all } s, t \in D, \qquad K = (2^{N+1} + 1)C_\gamma.$$

(f) On $\Omega^* = \{N < \infty\}$, $X|_D$ is uniformly continuous, so $Y_t := \lim_{D \ni r \to t} X_r$ exists for all $t \in [0, 1]$. Set $Y \equiv 0$ off Ω^*. Each Y_t is measurable (a limit along a sequence), and Y is γ-Hölder on $[0, 1]$ with constant K by continuity. For fixed t take $r_j \in D$, $r_j \to t$: then $X_{r_j} \to Y_t$ a.s. and $X_{r_j} \to X_t$ in probability by (b). Limits in probability are a.s. unique, so $Y_t = X_t$ a.s., and Y is a modification of X. Y does not depend on γ. Taking $\gamma_k \uparrow \beta/\alpha$ and intersecting the corresponding events $\Omega^*_{\gamma_k}$ shows that a.s. Y is γ-Hölder for every $\gamma < \beta/\alpha$.

2. Here t_0 is a local maximum if there is $\varepsilon > 0$ with $B_s \le B_{t_0}$ for all $s \in [0, \infty) \cap (t_0 - \varepsilon, t_0 + \varepsilon)$.

(a) $W_s = B_{t_0+s} - B_{t_0}$ is a BM. If t_0 is a local maximum, then $\max_{[0,1/n]} W \le 0$ for some n. By the reflection principle,

$$\mathbb{P}\big(\max_{[0,1/n]} W \ge a\big) = 2\mathbb{P}(W_{1/n} \ge a) \xrightarrow[a \downarrow 0]{} 1,$$

so $\mathbb{P}(\max_{[0,1/n]} W > 0) = 1$ for every n. Hence $\mathbb{P}\big(\bigcup_n \{\max_{[0,1/n]} W \le 0\}\big) = 0$, and $\{t_0 \text{ local max}\}$ is contained in a null set.

(b) B attains its maximum on $[0, 1]$ at some t^*. By (a) with $t_0 = 0$, a.s. $t^* \ne 0$. The process $W_s = B_1 - B_{1-s}$, $s \in [0, 1]$, is a continuous centred Gaussian process. Its covariance is

$$\operatorname{Cov}(W_s, W_u) = 1 - (1 - u) - (1 - s) + (1 - s \vee u) = s \wedge u,$$

so W is a BM on $[0, 1]$. Moreover $t^* = 1$ means $\min_{[0,1]} W \ge 0$, which has probability 0 by the argument of (a) applied to $-W$. So a.s. $t^* \in (0, 1)$, and t^* is a local maximum of B.

3.

(a) Let $Z_n = (B_{1/n} - B_{1/(n+1)})/\sqrt{1/n - 1/(n+1)}$. The intervals $[\frac{1}{n+1}, \frac{1}{n}]$ are disjoint, so the Z_n are i.i.d. $N(0,1)$. Then $\sum_n \mathbb{P}(Z_n \geq K) = \sum_n \mathbb{P}(Z_1 \geq K) = \infty$, and the second Borel–Cantelli lemma gives $\mathbb{P}(Z_n \geq K \text{ i.o.}) = 1$.

(b) Fix rationals $p < q$. The process $W_t = B_{p+t} - B_p$ is a BM. Applying (a) to W for every $K \in \mathbb{N}$, a.s. for all K there are infinitely many n with $1/n \leq q - p$ and

$$|W_{1/n} - W_{1/(n+1)}| \geq K|1/n - 1/(n+1)|^{1/2}.$$

So B is not $\frac{1}{2}$-Hölder on $[p,q]$. Intersecting over the countably many pairs (p,q), and using that every non-trivial interval contains some $[p,q]$, gives the claim.

4. Let $n \in \mathbb{N}$, $h = 1/n$, $t_k = kh$, $\delta = \varepsilon/4$. Let L be the piecewise linear interpolation of $(t_k, B_{t_k})_k$ and $Z = B - L$. On $[t_{k-1}, t_k]$,

$$Z_s = (B_s - B_{t_{k-1}}) - \tfrac{s-t_{k-1}}{h}(B_{t_k} - B_{t_{k-1}}), \qquad \text{so} \quad \sup_{[t_{k-1},t_k]} |Z| \leq 2 \sup_{u\in[0,h]} |B_{t_{k-1}+u} - B_{t_{k-1}}|.$$

Independence. $(Z_s)_s$ and $(B_{t_j})_j$ are jointly Gaussian and $\operatorname{Cov}(Z_s, B_{t_j}) = 0$. For $t_j \leq t_{k-1}$ this covariance equals $t_j - \frac{(t_k-s)t_j + (s-t_{k-1})t_j}{h} = 0$, and for $t_j \geq t_k$ it equals $s - \frac{(t_k-s)t_{k-1} + (s-t_{k-1})t_k}{h} = 0$. Hence $\sigma(Z_s : s \in [0,1])$ is independent of $(B_{t_1}, \dots, B_{t_n})$.

Let f_h be the linear interpolation of $(t_k, f(t_k))$. Then $|f_h - f| \leq \omega_f(h)$, where ω_f is the modulus of continuity. Consider the events

$$A = \{\max_k |B_{t_k} - f(t_k)| < \delta\}, \qquad C = \{\sup_{[0,1]} |Z| < \delta\}.$$

On $A \cap C$ we have $|B_t - f(t)| \leq |L_t - f_h(t)| + |f_h(t) - f(t)| + |Z_t| < 2\delta + \omega_f(h)$, which is $< \varepsilon$ once $\omega_f(h) < \varepsilon/2$.

$\mathbb{P}(A) > 0$, since $(B_{t_1}, \dots, B_{t_n})$ is an invertible linear image of i.i.d. $N(0,h)$ increments and so has an everywhere positive density. The suprema over the different intervals are independent (independent increments) and each has the law of $\sup_{[0,h]} |B|$. By scaling,

$$\mathbb{P}(C) \geq \mathbb{P}\big(\sup_{[0,h]} |B| < \delta/2\big)^n = \mathbb{P}\big(\sup_{[0,1]} |B| < \delta/(2\sqrt{h})\big)^n.$$

The inner probability tends to 1 as $h \to 0$, so it is positive for h small. Choosing such h with also $\omega_f(h) < \varepsilon/2$, independence gives $\mathbb{P}(\sup_{[0,1]} |B - f| < \varepsilon) \geq \mathbb{P}(A)\mathbb{P}(C) > 0$.

5.

(a) $X_0 = 0$ and X has continuous paths. For $0 \leq t_0 < \cdots < t_k$, the increments $X_{t_i} - X_{t_{i-1}} = a \cdot (B_{t_i} - B_{t_{i-1}})$ are functions of the independent vectors $B_{t_i} - B_{t_{i-1}}$, hence independent. Each is a sum of independent $N(0, a_j^2(t_i - t_{i-1}))$ variables, i.e. $N(0, |a|^2(t_i - t_{i-1})) = N(0, t_i - t_{i-1})$. So X is a BM.

(b) By (a) each coordinate $Y^i = a_i \cdot B$ is a BM, where a_i is the i-th row. All Y_t^i are linear combinations of the Gaussian family (B_t^j), so $(Y^1, \dots, Y^m)$ is jointly Gaussian with $\operatorname{Cov}(Y_s^i, Y_t^j) = (a_i \cdot a_j)(s \wedge t) = \delta_{ij}(s \wedge t)$. Uncorrelated jointly Gaussian families are independent: the finite-dimensional distributions factor, and a π–λ argument finishes the proof. So $Y^1, \dots, Y^m$ are independent BMs.

(c) For $R \in SO(d)$ the rows are orthonormal, so by (b) RB consists of d independent BMs, i.e. it is a d-dimensional BM. Since the law of a process is determined by its finite-dimensional distributions, $(RB_t)_t \overset{d}{=} (B_t)_t$.

6. For a partition π of $[0,T]$ write $Q_\pi(X) = \sum_i (X_{t_i} - X_{t_{i-1}})^2$. We use the known result that $Q_\pi(B) \to b - a$ in L^2 on any interval $[a,b]$; indeed $\mathbb{E}(Q_\pi(B) - (b-a))^2 = 2\sum \Delta t_i^2 \leq 2(b-a)|\pi|$.

(a) $Q_\pi(aB) = a^2 Q_\pi(B) \to a^2 T$.

(b) $W_t = c^{-1/2} B_{ct}$ is a BM, so $Q_\pi(B_{c\cdot}) = c Q_\pi(W) \to cT$.

(c) Let $Y_t = \int_0^t B_s\, ds$ and $M = \sup_{[0,T]} |B|$. Then $|Y_{t_i} - Y_{t_{i-1}}| \le M \Delta t_i$, so $Q_\pi(Y) \le M^2 T |\pi|$ and $\mathbb{E} Q_\pi(Y)^2 \le \mathbb{E}[M^4] T^2 |\pi|^2 \to 0$. Here $\mathbb{E} M^4 < \infty$, since $M \le \sup B + \sup(-B)$ and $\sup_{[0,T]} B \overset{d}{=} |B_T|$ by the reflection principle. The quadratic variation is 0.

(d) If $t_0 \in \pi$, then $Q_\pi(X) = Q_{\pi \cap [0,t_0]}(B) + 9\, Q_{\pi \cap [t_0,T]}(B) \to t_0 + 9(T - t_0)$ in L^2. Otherwise let $[t_{j-1}, t_j] \ni t_0$, $u = B_{t_0} - B_{t_{j-1}}$, $v = -3(B_{t_j} - B_{t_0})$. Then $Q_\pi(X) = Q_{\pi \cup \{t_0\}}(X) + 2uv$, and by independence $\mathbb{E}(2uv)^2 = 36(t_0 - t_{j-1})(t_j - t_0) \le 36|\pi|^2 \to 0$. Hence the quadratic variation is $t_0 + 9(T - t_0) = 9T - 8t_0$.